ORIENTATION, PHYSICS

CHAPTER 1

Measurements and Units

SI - system

- the base units are the building blocks (in total 7 base units)
- the derived units are formed by powers, products of quotients of the base units
- prefixes are used to express large or small numbers in simplier form.

Physical quantity = number * Unit

SI Base and Supplementary Units

Quantity	SI Unit Name	SI Unit Symbol
Base Unit		
Length	meter	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd
Supplementary Units		
Plane angle	radian	rad
Solid angle	steradian	sr

SI Prefixes

Factor	Prefix	Symbol
10^{24}	yotta	Y
10^{21}	zetta	Z
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{2}	hecto	h
10^{1}	deka	da
10^{0}	—	—
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a
10^{-21}	zepto	z
10^{-24}	yocto	y

Changing units

EXAMPLE Express $65 \frac{mi}{h}$ speed limit in meters per second.

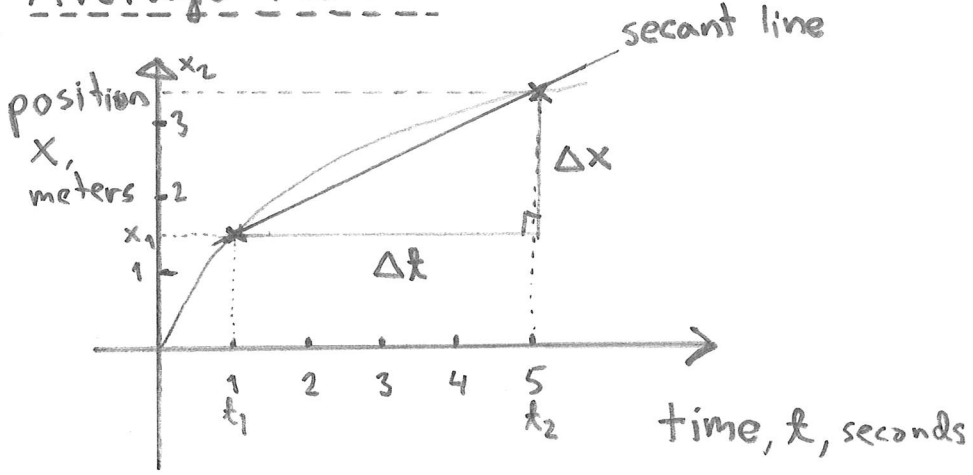
We know by the table $1 miles = 1609 m$ and $1 hour = 3600 s$.

Therefore $65 \frac{mi}{h} = 65 \cdot \frac{1609 m}{3600 s} \approx 29 \frac{m}{s}$

CHAPTER 2
Motion in a Straight Line

(= 1D kinematics)

Average motion



displacement $\Delta x = x_2 - x_1$

time $\Delta t = t_2 - t_1$

average velocity: $\bar{v}_{ave} = \frac{\Delta x}{\Delta t} = \frac{3,5-1,5}{5-1} = \frac{1}{2}\frac{m}{s}$

Observe:

geometrical interpretation

$\bar{v}_{ave}$ is the gradient of secant line in time-space coordinate system

$$\text{Average speed} = \frac{\text{distance travelled}}{\text{time interval}}$$

$$\text{Average velocity} = \frac{\text{displacement}}{\text{time interval}}$$

EXAMPLE To get a cheap flight from Houston to Kansas City (dist. 1000km), you have to connect in Minneapolis (700km north of Kansas City). The flight to Minneapolis takes 2,2h, then you have a 30-min layover, and then a 1.3h flight to Kansas City.

What are your average velocity and your average speed on this trip?



Time interval: 2,2h + 0,5h + 1,3h = 4h

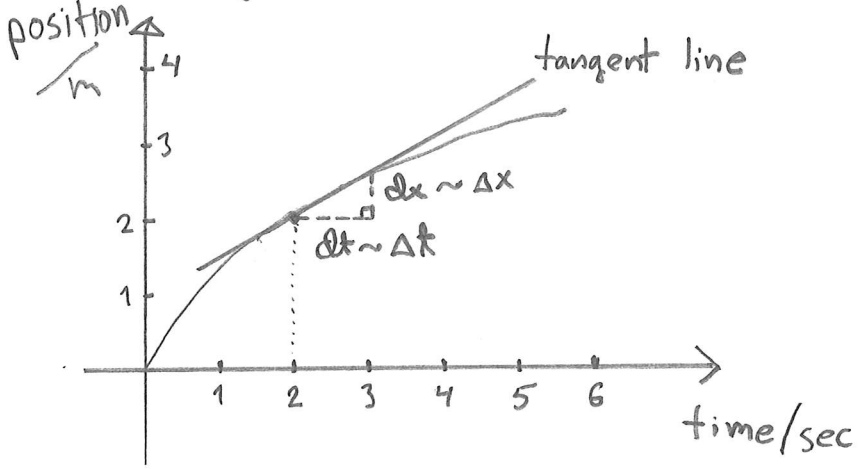
distance travelled: 1000km + 2×700km = 2400km

displacement: 1000km

average speed $= \frac{2400km}{4h} = 600\frac{km}{h}$, average velocity $= \frac{1000}{4} = 250\frac{km}{h}$

Instantaneous velocity

motion given in time – space (position) system



How to calculate the instantaneous velocity at $t = 2.0\,s$?

the problem is solved by graphical differentiation:

step 1 sketch a tangent line at $t = 2.0\,s$

step 2 determine the gradient of tangent

$$v(2) = \frac{dx}{dt} = \frac{0,5}{1} = 0,5\ \frac{m}{s}$$

Mathematically we understand this as a limit:

$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$ (derivative of x with resp to t)

the process is differentiation.

(In math we will study diff. in detail on period 3)

Leibniz notation ~ $\frac{dx}{dt}$

Newton notation ~ $x'(t)$

EXAMPLE The altitude of a rocket in the first half-minute of its ascent is given by $x = bt^2$, where $b = 2.90\ \frac{m}{s^2}$ (const).

Find a general expression for the rocket's velocity as a function of time and from it the instantaneous velocity at $t = 20\,s$.

Also find an expression for the average velocity.

④ solution by differentiation, inst velocity

$v(t) = x'(t) = \frac{dx}{dt} = 2bt = 2 \cdot 2.9 \cdot t = 5.8t$

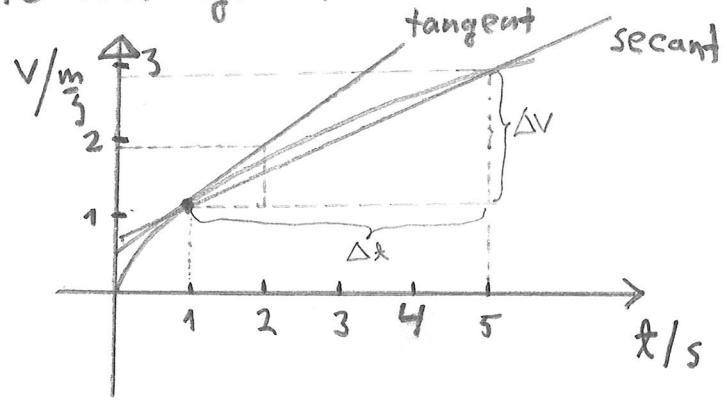
$v(20) = 5.8 \cdot 20 = 116 \frac{m}{s}$.

average velocity across $[0, t]$

$v_{ave} = \frac{\Delta x}{\Delta t} = \frac{bt^2 - 0}{t} = bt = 2{,}9t$ (half of inst. velocity)

Acceleration (velocity is changing)

motion given in time-velocity system



average acceleration ~ gradient of secant

instantaneous acceleration ~ gradient of tangent

Average acceleration: $\bar{a} = \frac{\Delta v}{\Delta t}$

Instantaneous acceleration: $a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$

Unit $[a] = \frac{m}{s^2}$

Travelled distance

How to calculate, when

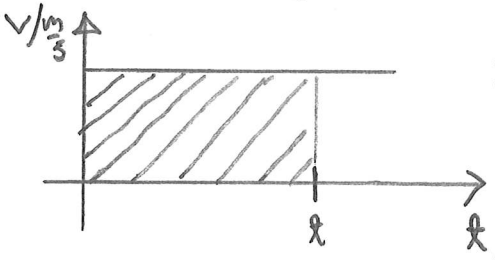
(a) velocity is constant?

(b) acceleration is constant?

(c) in general (i.e. $v(t)$ continuous)?

(a) If v constant, then $\boxed{x = v \cdot t}$

geometrical interpretation, when motion is given in time-velocity system



Shaded area $= v \cdot t = x$

(5)

(b) If acceleration is constant, then

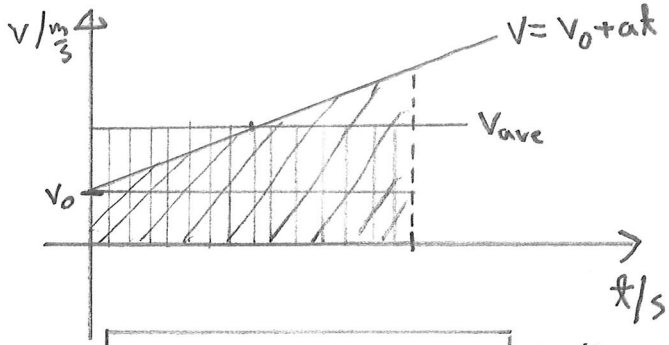
$$\boxed{v(t) = v_0 + at}$$

average velocity across $[0,t]$:

$$v_{ave} = \frac{v_0+v}{2} = \frac{v_0+v_0+at}{2} = v_0 + \frac{1}{2}at$$

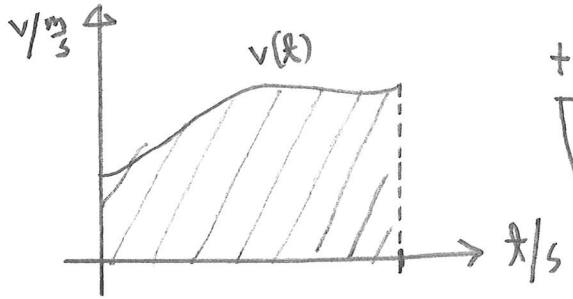
and

$$\boxed{x = v_{ave} \cdot t = v_0 \cdot t + \frac{1}{2}at^2}$$



$\boxed{x = \text{shaded area}}$ half parallellogram or rectangle

(c) If $v(t)$ is continuous function (general case)



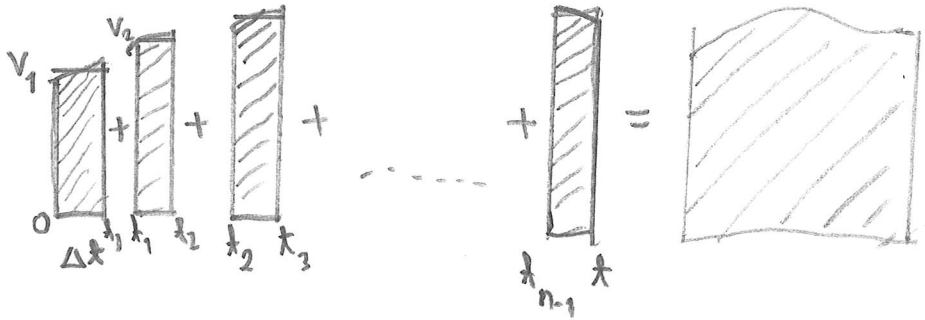
travelled distance:

$\boxed{x \text{ is the shaded area}}$

How to understand this?

We split the interval $[0,t]$ to small pieces $[0,t_1],[t_1,t_2],\dots,[t_n,t]$. where Δt $t_i - t_{i-1}$ is const

In each interval, average velocity is V_i and travelled distance is the area



Total distance is the sum:

$$v_1\Delta t + v_2\Delta t + \dots + v_n\Delta t = \sum_{i=1}^{n} v_i \Delta t \xrightarrow[\Delta t]{} x = \int_0^t v(t)\,dt$$

This method is called graphical integration.

Formulas for constant acceleration:

$$\boxed{v = at + v_0} \Rightarrow t = \frac{v - v_0}{a}$$

$$\boxed{x = v_0 t + \frac{1}{2}at^2} \Rightarrow x = t\left(v_0 + \frac{1}{2}at\right)$$

from these equations, we obtain

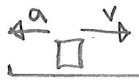
$$x = \frac{v - v_0}{a}\left(v_0 + \frac{1}{2}a\frac{(v - v_0)}{a}\right) = \frac{1}{2}\frac{(v - v_0)(v + v_0)}{a}$$

which simplifies to

$$x = \frac{1}{2a}\left(v^2 - v_0^2\right)$$

or $$\boxed{v^2 - v_0^2 = 2ax}$$

EXAMPLE A Jetliner touches down at $270\,\frac{km}{h}$. The plane then decelerates at $4{,}5\,\frac{m}{s^2}$ (i.e. $a = -4{,}5\,\frac{m}{s^2}$). What's the minimum runaway length on which this aircraft can land?

Solution: 

acceleration $a = -4{,}5\,\frac{m}{s^2}$ is constant, so we can use the formula $v^2 - v_0^2 = 2ax$.

Now $v = 0$, $v_0 = 270\,\frac{km}{h} = 270 \cdot \frac{1000\,m}{3600\,s} = 75\,\frac{m}{s}$

We get: $x = \frac{v^2 - v_0^2}{2 \cdot a} = \frac{-75^2}{-2 \cdot 4{,}5} = \boxed{625\,m}$

EXAMPLE A speeding motorist zooms through a $50\,\frac{km}{h}$ zone at $75\,\frac{km}{h}$ without noticing a stationary police car. The police officer immediately heads after the speeder at $a = 2{,}5\,\frac{m}{s^2}$. When the officer catches up to the speeder, how far down the road are they, and how fast is the police car going?

Solution Motorist: constant velocity

$$x = v \cdot t$$

Police: constant acceleration

$$x = \underbrace{v_0}_{=0} t + \frac{1}{2}at^2 = \frac{1}{2}at^2$$

From these equations, they have same x at

(7)

$$\frac{1}{2}at^2 = v\cdot t \quad \Rightarrow t\left(\frac{1}{2}at - v\right) = 0$$

$$\Rightarrow \begin{cases} t = 0 \\ t = \frac{2v}{a} = \frac{2\cdot 75\,\frac{1000m}{3600s}}{2,5} = \boxed{16,7s} \end{cases}$$

At this moment

$$x = \frac{1}{2}a\cdot t^2 = \frac{1}{2}\cdot 2,5\cdot\left(\frac{50}{3}\right)^2 = 347m \approx \boxed{350m}$$

The velocity of the police car

$$v = a\cdot t = 2,5\cdot\frac{50}{3} = 41,67\,\frac{m}{s} = \boxed{150\,\frac{km}{h}}$$

The acceleration of Gravity



$\vec{g}$ near Earth's surface

$$g \approx 9,8\,\frac{m}{s^2}$$

EXAMPLE A diver drops from a 10-m-high cliff. A what speed does he enter the water, and how long is he in the air?

Solution

acceleration is constant $g = 9,8\,\frac{m}{s^2}$

Therefore we can use the equation

$v^2 - v_0^2 = 2gx$, where $v_0 = 0$, $x = 10m$

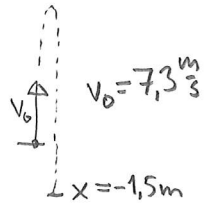
$$\Rightarrow v = \sqrt{2\cdot 9,8\cdot 10} = \boxed{14\,\frac{m}{s}}$$

And $v = gt \Rightarrow t = \frac{v}{g} = \frac{14}{9,8} \approx \boxed{1,4s}$

EXAMPLE You toss a ball straight up at $7,3\,\frac{m}{s}$; it leaves your hand at 1,5m above the floor. Find when it hits the floor, the maximum height and its speed when it passes your hand on the way down?

Solution

Acceleration is constant $g = -9,8\,\frac{m}{s^2}$



$$x = v_0 t + \frac{1}{2}gt^2 \Leftrightarrow +\frac{1}{2}gt^2 + v_0 t - x = 0$$

using the quadratic formula

$$-4,9t^2 + 7,3t + 1,5 = 0, \quad t = \frac{-7,3 \pm \sqrt{7,3^2 + 4\cdot 4,9\cdot 1,5}}{-2\cdot 4,9} = 1,673s$$

$$\approx \boxed{1,7s}$$

(8)

What is the maximum height?

At highest point $v=0$, so we get

$$V = v_0 + gt \Rightarrow t = \frac{0-7,3}{-9,8} = \frac{7,3}{9,8}\,s \approx 0,7449\,s$$

and the height

$$h = v_0 \cdot t + \frac{1}{2}gt^2 = 7,3 \cdot \frac{7,3}{9,8} - \frac{1}{2} \cdot 9,8 \cdot \left(\frac{7,3}{9,8}\right)^2 \approx 2,72\,m$$

The height from the floor: $1,5 + 2,7 \approx \boxed{4,2\,m}$

When a ball is back at $x=0$?

$$0 = v_0 t + \frac{1}{2}gt^2 \Rightarrow t = \frac{-2v_0}{g}$$

and

$$v = v_0 + gt = v_0 - \frac{2v_0}{g} \cdot g = -v_0 = \boxed{-7,3\,\frac{m}{s}}$$

the same speed as v_0 (but downwards)

CHAPTER 3

VECTORS

Will be studied in detailed in Math

Scalar quantity = number x phys unit

Vector quantity = number x direction x phys unit

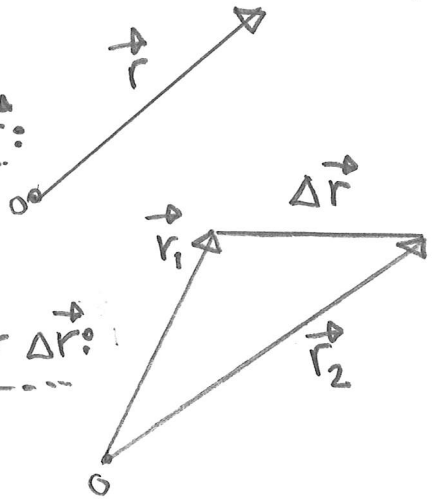
EXAMPLE

Position vector $\vec{r}$:

Displacement vector $\Delta\vec{r}$:

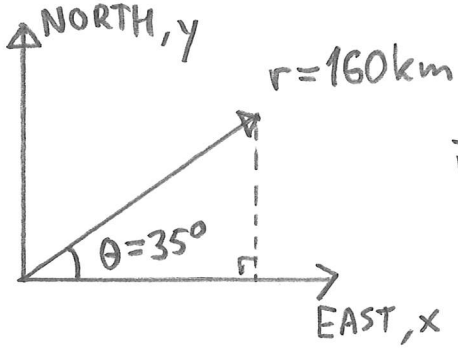
initial position $\vec{r}_1$

final position $\vec{r}_2$



$$\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$$

EXAMPLE You drive to a city 160km from home, going 35° north east. Express your new position in unit vector notation.



$$\vec{r} = 160\cdot\cos(35^\circ)\vec{i} + 160\cdot\sin(35^\circ)\vec{j}$$

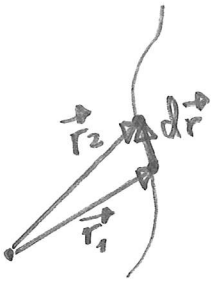
$$\vec{r} = 131\,\text{km}\,\vec{i} + 92\,\text{km}\,\vec{j}$$

Velocity and Acceleration vectors

average velocity vector: $\vec{v}_{ave} = \dfrac{\Delta\vec{r}}{\Delta t}$

instantaneous velocity vector:

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta\vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$



we can write this in the form

$$\vec{v} = \frac{dx}{dt}\vec{i} + \frac{dy}{dt}\vec{j} = v_x\vec{i} + v_y\vec{j}.$$

average acceleration vector:

$$\vec{a}_{ave} = \frac{\Delta\vec{v}}{\Delta t}$$

instantaneous acceleration vector:

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta\vec{v}}{\Delta t} = \frac{d\vec{v}}{dt}$$

which can be written as

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\vec{i} + \frac{dv_y}{dt}\vec{j} = a_x\vec{i} + a_y\vec{j}$$

Relative Motion

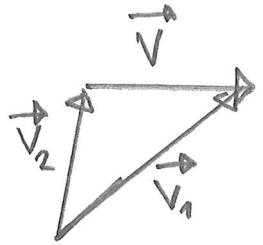
airplane is flying with velocity

- $\vec{V}_1$ relative to the ground
- $\vec{V}_2$ relative to the air

wind is $\vec{V}$

then:

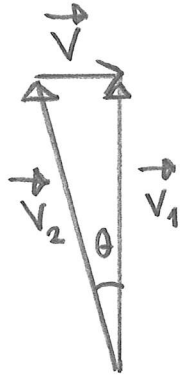
$$\vec{V}_1 = \vec{V}_2 + \vec{V}$$



EXAMPLE A Jetliner flies at $960\ \frac{km}{h}$ relative to the air. It's going from Houston to Omaha, 1290 km northward. At cruising altitude a wind is blowing eastward at $190\ \frac{km}{h}$.

(a) In what direction should the plane fly?

(b) How long will the trip take?



We have: $V = 190\ \frac{km}{h}$, $\vec{V} = 190\vec{i}$

$V_2 = 960\ \frac{km}{h}$, $\vec{V}_1 = V_1\vec{j}$

By the figure: $\sin\theta = \frac{190}{960}$

therefore: $\theta = \sin^{-1}\left(\frac{190}{960}\right) \approx \boxed{11.4^\circ}$

And $v_1 = 960 \frac{km}{h} \cdot \cos(11.4°)$

Because $s = v_1 \cdot t$

$$\Rightarrow \quad t = \frac{s}{v_1} = \frac{1290\,km}{960 \frac{km}{h} \cos(11.4°)} \approx \boxed{1{,}37h}$$

$$\approx \boxed{1h\,22min\,15s}$$

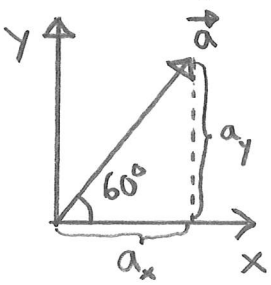
Constant acceleration

We can extend 1-dim formulas to higher dimensions. For a constant acceleration:

$$\boxed{\vec{v} = \vec{v}_0 + \vec{a}t}$$

$$\boxed{\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2}$$

EXAMPLE You are windsurfing at $7{,}3 \frac{m}{s}$ when a gust hits, accelerating your sailboard at $0.82 \frac{m}{s^2}$ at 60° to your original direction. If the gust lasts 8.7s, what is the boards displacement during this time?



$$\vec{v}_0 = \underbrace{7{,}3}_{v_{x,0}}\vec{i}$$

constant acceleration

$$\vec{a} = \underbrace{0.82 \cdot \cos 60°}_{a_x}\vec{i} + \underbrace{0.82 \cdot \sin 60°}_{a_y}\vec{j}$$

Displacement:

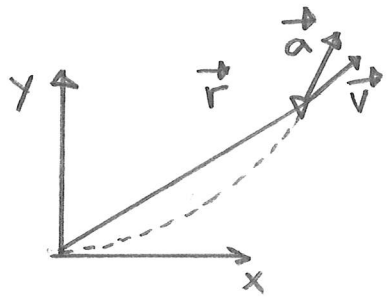
write components of $\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2}\vec{a}t^2$

$$\Rightarrow \begin{cases} x = x_0 + v_{x,0} t + \frac{1}{2} a_x t^2 \\ y = y_0 + v_{y,0} t + \frac{1}{2} a_y t^2 \end{cases}$$

When $t = 8.7s$, we have

$$x = 0 + 7{,}3 \cdot 8.7 + \frac{1}{2} \cdot 0{,}82 \cdot \cos 60° \cdot 8.7^2 = \boxed{79.0m}$$

$$y = 0 + 0 \cdot 8.7 + \frac{1}{2} \cdot 0.82 \cdot \sin 60° \cdot 8.7^2 = \boxed{26.9m}$$

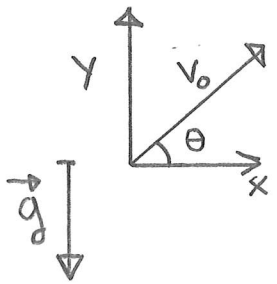


By Pythagoras:

$$r = \sqrt{79^2 + 26.9^2} \approx \boxed{83m}$$

Projectile Motion

A projectile is an object that is launched into the air and then moves predominantly under the influence of gravity.



assumptions:

(1) gravitational acceleration is constant $\vec{g} = -9.81 \frac{m}{s^2} \hat{j}$

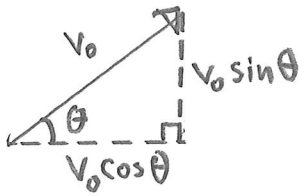
(2) neglect air resistance

Problem: Solve the motion, i.e. find v_x, v_y, x and y in terms of v_0, θ, g and t.

IDEA: We know that:

(1) in x-direction velocity is constant

(2) in y-direction acceleration is constant



By the figure and graphical integration:

$$v_x = \text{constant} = v_{x,0} = v_0 \cdot \cos\theta$$

$$x = x_0 + v_0 \cdot \cos\theta \cdot t$$

$$a_y = \text{constant} = -g$$

$$v_y = v_{y,0} - g \cdot t = v_0 \sin\theta - g \cdot t$$

$$y = y_0 + v_0 \cdot \sin\theta \cdot t - \frac{1}{2} g t^2$$

We can choose $x_0 = 0$, $y_0 = 0$,

Therefore the solution of projectile motion:

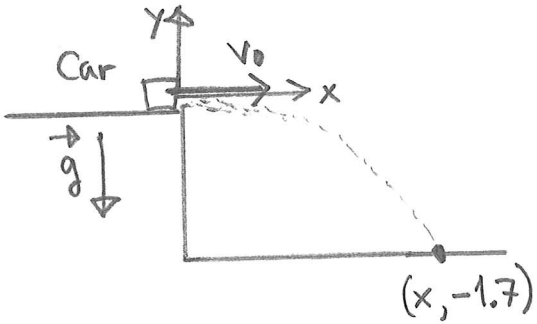
$$v_x = v_0 \cdot \cos(\theta)$$

$$v_y = v_0 \cdot \sin(\theta) - g \cdot t$$

$$x = v_0 \cdot \cos(\theta) \cdot t$$

$$y = v_0 \cdot \sin(\theta) \cdot t - \frac{1}{2} g t^2$$

EXAMPLE A raging flood has washed away a section of highway, creating a gash 1.7 m deep. A car moving at $31 \frac{m}{s}$ goes straight over the edge. How far from the edge of the washout does it land?



$v_0 = 31 \frac{m}{s}$

$\theta = 0$

$g = 9.81 \frac{m}{s^2}$

Projectile motion:

$$\begin{cases} v_x = 31 \cdot \cos 0 = 31 \\ v_y = 31 \cdot \sin 0 - 9.81 \cdot t = -9.81 \cdot t \\ x = 31 \cdot t \\ y = -\frac{1}{2} \cdot 9.81 \cdot t^2 \end{cases}$$

At the final point, we have a condition

$$y = -1,7.$$

Using this, we can solve the time of flight:

$$-\frac{1}{2} \cdot 9.81 \cdot t^2 = -1,7$$

$$\Rightarrow t = (\pm)\sqrt{\frac{2 \cdot 1.7}{9.81}} \approx 0.5887 s$$

Then we can solve x:

$$x = 31 \cdot t = 31 \cdot 0.5887 \approx 18.25 m \approx \boxed{18 m}$$

Projectile Trajectories

Projectile trajectory is a parabola:

By elimination: $x = v_0 \cdot \cos\theta \cdot t \Rightarrow t = \frac{x}{v_0 \cos\theta}$

substitute to y-equation,

We get $y = v_0 \sin\theta \cdot \frac{x}{v_0 \cos\theta} - \frac{1}{2} g \cdot \frac{x^2}{v_0^2 \cos^2\theta}$

or

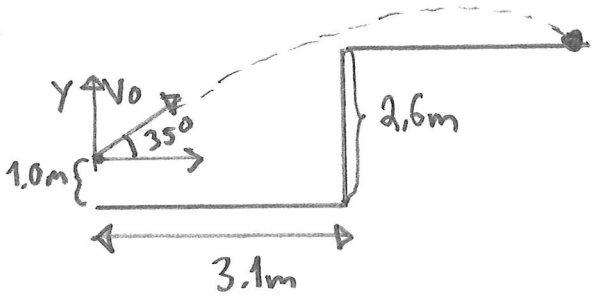
$$\boxed{y = \tan\theta \cdot x - \frac{g}{2 v_0^2 \cos^2\theta} \cdot x^2}$$

Clearly a parabola.

EXAMPLE A worker stands in a 2.6 m deep hole, 3.1 m from the edge of the hole. He tosses a hammer to a companion outside the hole. If the hammer leaves his hand 1.0 m above the bottom of the hole at an angle 35°,

(a) What is the minimum speed it needs to clear the edge of the hole?

(b) How far from the edge of the hole does in land?



(a) A trajectory is a parabola, which passes through a point (3.1, 1.6)

note: 1.6 = 2.6 − 1

so, we obtain

$$1.6 = \tan 35° \cdot 3.1 - \frac{9.81}{2 \cdot v_0^2 \cdot \cos^2 35°} \cdot 3.1^2$$

solving this

$$v_0 = \sqrt{\frac{9.81 \cdot 3.1^2}{2 \cdot (\tan(35°) \cdot 3.1 - 1.6) \cdot \cos^2(35°)}} \approx 11 \frac{m}{s}$$

(b) Find first the time, when $y = 1.6$.

Using y-equation:

$$y = v_0 \cdot \sin\theta \cdot t - \frac{1}{2} g t^2$$

$$\Leftrightarrow \quad \frac{1}{2} \cdot 9.81 \cdot t^2 - 11 \cdot \sin(35°) \cdot t + 1.6 = 0$$

By the quadratic formula

$$t = \frac{11 \sin 35° \pm \sqrt{(11 \cdot \sin 35°)^2 - 4 \cdot \frac{1}{2} \cdot 9.81 \cdot 1.6}}{2 \cdot \frac{1}{2} \cdot 9.81} \approx 0.9564$$

Note: we choose $\pm$, for the latter time at 1.6m height.

And then we can calculate x coordinate:

$$x = v_0 \cos\theta \cdot t = 11 \cdot \cos(35^\circ) \cdot 0.9564 \approx 8.7\,m$$

From the edge $= 8{,}7\,m - 3{,}1\,m \approx \boxed{5.6\,m}$

The Range of a Projectile

What is horizontal range for a projectile?

If wet set $y=0$, in the projectile trajectory (equation), we obtain

$$0 = \tan\theta \cdot x - \frac{g}{2v_0^2\cos^2\theta} \cdot x^2 .$$

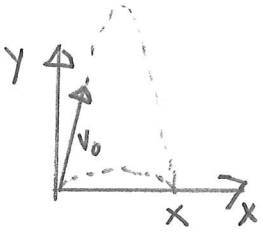
There are two solutions:

$$x = 0 \quad \text{or} \quad x = \frac{2v_0^2\cos^2\theta \cdot \tan\theta}{g} = \frac{2v_0^2}{g} \cdot \sin\theta\cos\theta .$$

Because $\sin 2\theta = 2\sin\theta\cos\theta$,

we get $\boxed{x = \frac{v_0^2}{g} \cdot \sin(2\theta)}$

EXAMPLE After a short engine firing, an atmosphere-probing rocket reaches $4{,}6\ \frac{km}{s}$. If the rocket must land within 50km of its launch site, what is the maximum allowable deviation from a vertical trajectory?



Requirement $0 \le x \le 50\,km$

Using the equation $x = \frac{v_0^2}{g}\sin(2\theta)$

we obtain $\sin(2\theta) = \frac{g \cdot x}{v_0^2} = \frac{9.81\,\frac{m}{s^2} \cdot 50\,km}{\left(4{,}6\,\frac{km}{s}\right)^2}$

has two solutions

$$2\theta = \sin^{-1}\left(\frac{9.81 \cdot 50 \cdot 10^3}{(4.6 \cdot 10^3)^2}\right) \Rightarrow \theta_1 = 0.66^\circ$$

or

$$2\theta = 180^\circ - \sin^{-1}\left(\frac{9.81 \cdot 50 \cdot 10^3}{(4.6 \cdot 10^3)^2}\right) \Rightarrow \theta_2 = 89.34^\circ$$

The deviation from vertical: $90 - \theta_2 \approx \boxed{0.66^\circ}$
(2nd angle θ_2)